U.S. Healthcare Payer API & RCM Automation Feasibility Report

Claim Status | Eligibility | Benefits | Patient Responsibility | Prior Authorization | Denials | Appeals | Payer Intelligence

PREPARED FOR	**AI Revenue Recovery OS**
DATE	August 18, 2026
SCOPE	Evidence-backed payer / API capability database — what RCM data can be obtained electronically today, through which channel, at what cost, and where portal / voice / human fallback remains necessary.
CLASSIFICATION	Confidential · Investor & product/engineering feasibility

Demo / research disclosure

This is an analytical feasibility database compiled from official CMS, X12, HL7, CAQH and vendor documentation (research completed August 2026). Capability values are marked YES / PARTIAL / NO / UNKNOWN; 'free sandbox' is never equated with 'free production', and 'direct payer API' is never equated with 'clearinghouse connectivity'. The 43-organization payer set is representative, not exhaustive. No PHI is used.

Contents

1. Executive Summary

The U.S. payer ecosystem is **not API-uniform** — but it is far more electronically accessible than a first glance suggests. The standardized transactions that underpin revenue-cycle work — eligibility (270/271), claim status (276/277), claims (837) and remittance/denials (835) — are HIPAA-mandated, highly adopted, and reachable **today** through clearinghouse APIs that offer free sandboxes and, in at least one case, a free production tier. What is scarce is **direct, free, provider-facing payer APIs**: those remain the exception, and most payer FHIR portals expose patient-authorized member data, not provider RCM feeds.

	Standard	API	Free Sandbox	Prod API	Portal	Voice
Claim Status	YES	YES	YES	YES	YES	YES
Eligibility	YES	YES	YES	YES	YES	PARTIAL
Benefits	PARTIAL	YES	YES	YES	YES	PARTIAL
Patient Responsibility	PARTIAL	PARTIAL	PARTIAL	PARTIAL	YES	PARTIAL
Prior Authorization	YES	PARTIAL	PARTIAL	PARTIAL	YES	YES
Denials (835/ERA)	YES	YES	YES	YES	NO	NO
Appeals	NO	PARTIAL	NO	PARTIAL	YES	YES

Figure 1. RCM capability heatmap — standard, API, free sandbox, production API, portal and voice fallback by workflow. Green = YES, amber = PARTIAL, red = NO.

The eleven questions this report answers

Question	Answer
Can we obtain these RCM data points electronically?	Yes for eligibility, benefits, claim status, claims and denials/ERA; partially for prior authorization and patient responsibility; largely no (non-standard) for appeals.
Which have standardized transactions?	Eligibility 270/271, claim status 276/277, prior auth 278, claims 837, remittance 835 — all HIPAA-mandated at ASC X12 005010. Appeals have no standard.
Which have APIs?	All of the above are exposed as JSON/REST APIs by clearinghouses (Stedi, Optum, Availity). Direct payer REST APIs for RCM are rare.
Which APIs are free?	Free sandboxes are common (Stedi, Optum, Availity-mock). A free production tier is rare — Stedi offers 100 free transactions/month. Production is otherwise usage-priced or contracted.
Which require payment?	Production transactions at Optum, Availity, Waystar are contracted/enterprise; Stedi is pay-as-you-go with no minimum.
Which require enrollment?	835/ERA always requires per-payer enrollment; eligibility usually requires none; claims and PA vary by payer.
Which payers have direct APIs?	For provider RCM, essentially only Optum (a clearinghouse owned by UnitedHealth Group). Other payers publish CMS-9115 patient-access FHIR, not provider RCM APIs.
Where can we use clearinghouse APIs?	Everywhere for eligibility, benefits, claim status, claims and 835 — clearinghouse networks reach 3,500–5,000+ payers.

Question	Answer
Where are portals still necessary?	Prior authorization, appeals, document upload, and payer-specific detail not surfaced electronically.
Where is AI voice still necessary?	When digital channels cannot resolve status, obtain a reference, or complete an authorization/appeal — the last resort after EDI, API/FHIR and portal.
What can we realistically build in V1?	Eligibility, benefit verification and claim status via clearinghouse API, plus 835-based denial intelligence and a patient-responsibility estimation engine.

Investor-ready conclusion

The U.S. payer ecosystem is not API-uniform. However, significant portions of eligibility, benefits, claims, claim status and remittance workflows already have standardized electronic transactions and API infrastructure. Prior authorization is moving toward FHIR-based APIs for CMS-regulated payers (Jan 1, 2027). The opportunity is therefore **not to build a new payer network from scratch, but to build an intelligent orchestration layer** over existing EDI, clearinghouse, payer API, FHIR, portal and voice infrastructure — with a proprietary Payer Intelligence layer as the durable moat.

2. Research Methodology

Findings were compiled from official and primary sources first: CMS (rules, fact sheets, Administrative Simplification), ASC X12 (transaction standards and code lists), HL7 (FHIR / Da Vinci and CARIN implementation guides), CAQH (CORE operating rules and the CAQH Index), and vendor developer documentation (Stedi, Optum/Change Healthcare, Availity, Waystar, Zelis and others). Trade press was used only to corroborate figures, never as a primary claim.

- **Evidence discipline.** Each material claim is tied to a source. A payer homepage was never treated as evidence of an API capability; the source had to actually document the capability.
- **No forced binaries.** Values are YES / PARTIAL / NO / UNKNOWN / NOT APPLICABLE. PARTIAL is used where only some plans, states, products or transactions are supported.
- **Critical distinctions preserved.** Standard vs. electronic transaction vs. API vs. public API vs. free API vs. free sandbox vs. paid production; and direct payer API vs. clearinghouse connectivity.
- **Confidence levels.** High = official payer/CMS/HL7/X12/clearinghouse docs; Medium = reputable vendor/tech source; Low = third-party directory (never presented as confirmed).
- **Coverage limitation.** The U.S. payer ecosystem contains thousands of payer IDs. The 43-organization set here is representative across categories, not exhaustive; gaps are listed explicitly in Section 20 and the Research Gaps appendix.

Analytical scores. The Automation Readiness Score (0–100) and Digital Coverage Score are *our* weighted analytical constructs, not industry-standard measures. The readiness rubric weights digital API/FHIR (25), standard EDI (15), real-time (10), portal (10), prior-auth digital (10), denial electronic data (10), appeal electronic (5), documentation (5), sandbox (5) and a clear enrollment path (5).

3. The U.S. Payer Ecosystem

The database models three distinct layers: the **Payer Organization** (e.g., Elevance Health), the **Payer Plan / Product** (commercial, Medicare Advantage, Medicaid managed care, exchange), and the **Payer ID** (clearinghouse-specific routing identifiers). A single organization often spans many products and hundreds of payer IDs; a parent organization's technology does not necessarily apply to every subsidiary, so capabilities are assessed per organization with that caveat.

Payer category	Count	Examples	CMS-0057-F impacted?
National commercial	11	UnitedHealthcare, Aetna, Cigna, Humana	Partial (Medicare/Medicaid lines)
Blue Cross Blue Shield	14	HCSC, Highmark, Florida Blue, BSC	Partial (by product)
Medicaid/CHIP MCO	6	CareSource, AmeriHealth Caritas	Yes
Medicare Advantage	1	WellCare	Yes
Marketplace / QHP	2	Oscar, Ambetter	Yes (FFE QHPs)
Third-party administrator	5	UMR, Meritain, GEHA	No (commercial/ERISA)
Government FFS	4	Medicare FFS, Medicaid FFS, TRICARE	Medicaid: yes

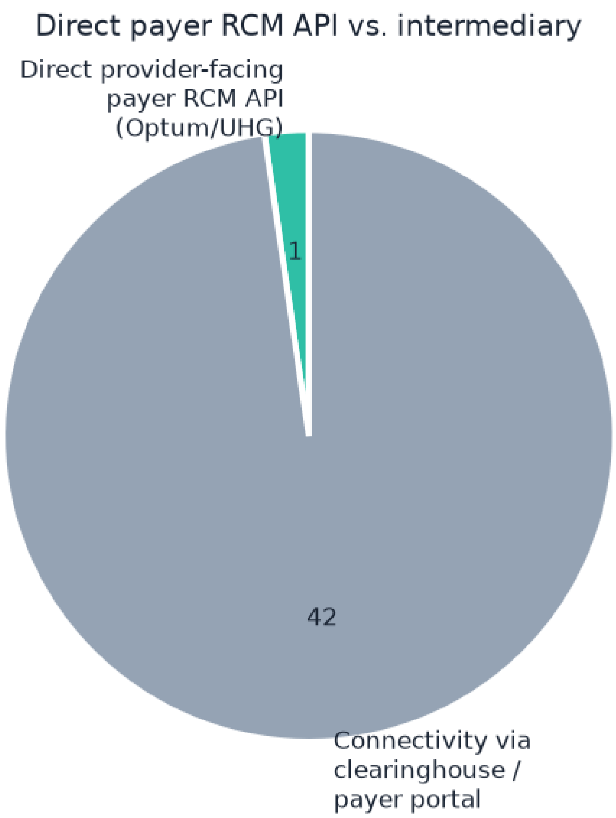

Figure 2. For provider-facing RCM, direct payer APIs are the rare exception; connectivity runs through clearinghouses and payer portals.

4. Healthcare Transaction Standards

HIPAA (45 CFR Part 162) requires covered entities to use adopted ASC X12 standards (version 005010) when conducting these transactions electronically. These mandates are the foundation of RCM automation: because payers *must* support them, a clearinghouse can reach virtually any payer for eligibility, status, claims and remittance.

X12	Transaction	HIPAA	Version	RCM use
270/271	Eligibility & Benefit Inquiry / Response	Mandated	005010X279A1	Coverage, plan, benefits, patient financial responsibility
276/277	Claim Status Request / Response	Mandated	005010X212	Real-time claim status
278	Health Care Services Review (Referral / Prior Authorization)	Mandated	005010X217	Prior authorization request/response
837P/I/D	Health Care Claim (Professional / Institutional / Dental)	Mandated	005010X222A1 / X223A2 / X224A2	Claim submission
835	Health Care Claim Payment/Advice (ERA)	Mandated	005010X221A1	Remittance, payment, denial (CARC/RARC)
834	Benefit Enrollment and Maintenance	Mandated	005010X220A1	Enrollment
820	Premium Payment	Mandated	005010X218	Premium payment
277CA	Claim Acknowledgment	Not mandated (widely used)	005010X214	Pre-adjudication accept/reject of 837
999	Implementation Acknowledgment	Not mandated (industry)	005010X231A1	Syntax acknowledgment
TA1	Interchange Acknowledgment	Not mandated	X12 interchange	Envelope/delivery ack

4.1 Denial codes (carried in the 835)

The 835 ERA carries denial detail as **CARC** (Claim Adjustment Reason Codes, maintained by X12) and **RARC** (Remittance Advice Remark Codes, maintained by CMS). CAQH CORE maintains the federally required CARC/RARC code combinations. Because 835 is ~97% electronic, denial data is fully machine-readable — the basis for AI root-cause and recovery analysis.

4.2 Operating rules

CAQH CORE Operating Rules (HIPAA-adopted under ACA §1104) mandate richer 271 benefit content (copay, coinsurance, deductible, remaining deductible for defined service types), real-time response and availability requirements, connectivity 'safe harbor', and 835/EFT reassociation via the TRN trace number in the Nacha CCD+ addenda.

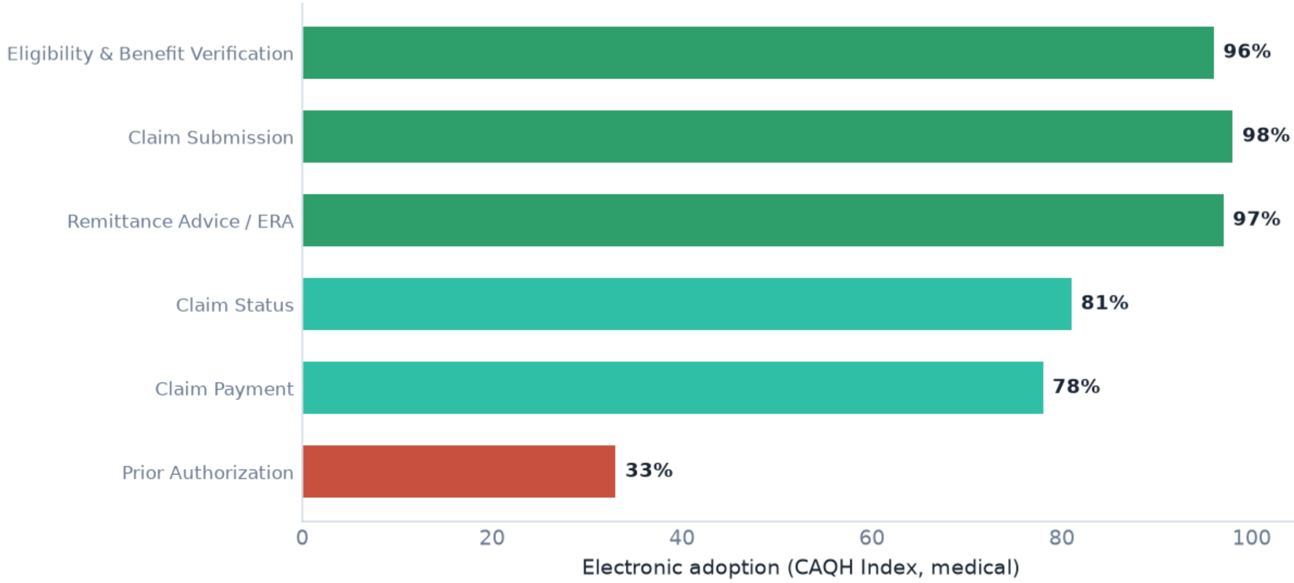

Figure 3. Electronic adoption by transaction (CAQH Index). Prior authorization is the clear laggard. Adoption figures: CAQH Index (2023–2024 data years). Manual transactions cost on average ~$5.43 more than electronic; CAQH estimates $20B+ in remaining annual automation savings and ~$222B avoided through existing automation.

5. Claim Status

Claim status uses the HIPAA-mandated 276/277 pair and is ~81% electronic — strong, with a real remaining automation gap. Real-time status is available today through clearinghouse APIs (Stedi, Optum, Availity) returning JSON. Direct payer claim-status APIs are UNKNOWN except Optum. Where a payer returns insufficient status electronically, the platform escalates to portal, then AI voice.

> **Build status: GREEN**
>
> 276/277 real-time via clearinghouse APIs today; 81% electronic. Build first.

6. Eligibility Verification

Eligibility (270/271) is the most-adopted transaction at ~96% electronic and is the strongest V1 candidate. Clearinghouse APIs return active coverage, coverage dates, member/plan status and network indicators in real time, with free sandboxes and (Stedi) a free production tier. Enrollment is usually not required for eligibility. This is a build-first capability.

Build status: GREEN

270/271 is 96% electronic and available now via clearinghouse APIs with free sandboxes (Stedi, Optum). Build first.

7. Benefit Verification

Benefit detail rides on the CORE-enhanced 271 response. Copay, coinsurance and deductible are reliably present; remaining deductible, OOP maximum/remaining, service-specific and network benefits, and real-time accumulators are payer-variable — hence PARTIAL. Do not assume every 271 contains every field; test empirically per payer.

Benefit field	Availability via 271
Active coverage / dates	YES
Plan / product info	YES
Copay	YES
Coinsurance	YES
Deductible	YES
Deductible remaining	PARTIAL
Out-of-pocket maximum	PARTIAL
OOP remaining	PARTIAL
Service-specific benefits	PARTIAL
Network / non-network benefits	PARTIAL
Coordination of benefits	PARTIAL
Authorization indicators	PARTIAL
Allowed amount	UNKNOWN / rarely in 271
Accumulators (real-time)	PARTIAL

Build status: GREEN

Structured benefits parsed from CORE-enhanced 271 (copay, coinsurance, deductible, remaining). Some fields payer-variable → PARTIAL on completeness.

8. Patient Responsibility Estimation

This is **not** a single API lookup. Payer 271 (and accumulator data where available) supplies deductible, remaining deductible, copay, coinsurance and OOP figures; but the **allowed amount** is rarely in the 271, and exact liability depends on contract rates, accumulator timing and claim adjudication. Patient responsibility must therefore be **calculated by our own estimation engine** and presented as an *estimate*, never as a guaranteed liability. Separate 'payer-provided data' from 'our calculated estimate' throughout the product.

Build status: YELLOW

Payer 271 supplies deductible/coinsurance/copay/accumulators, but exact liability must be CALCULATED by our estimation engine using allowed amounts + accumulators. Label 'Estimated'.

9. Prior Authorization

Prior authorization is the least-automated workflow. The 278 standard exists and is HIPAA-mandated, but only ~one-third of PA is fully electronic — the transaction historically cannot carry the clinical documentation payers require, so portals and fax dominate. The trajectory is FHIR: HL7 Da Vinci CRD (is PA required?), DTR (gather documentation), and PAS (submit/receive decision). Under CMS-0057-F, CMS-regulated payers must stand up a FHIR Prior Authorization API by **January 1, 2027**, with operational PA rules (72-hour expedited / 7-day standard decisions; specific denial reasons; public metrics) beginning **January 1, 2026**. CMS gives enforcement discretion for all-FHIR PAS implementations but does not ban the X12 278.

Da Vinci IG	Name	Role
CRD	Coverage Requirements Discovery	Surfaces coverage rules & whether PA is required (CDS Hooks)
DTR	Documentation Templates and Rules	Runs payer questionnaires/CQL in the EHR to gather PA documentation
PAS	Prior Authorization Support	Submits FHIR PA request, returns decision; bridges to X12 278
CDex	Clinical Data Exchange	Provider↔payer clinical data & attachments for claims/PA
PDex	Payer Data Exchange	Payer-held member data (payer-to-payer, provider access, patient access)
CARIN-BB	CARIN IG for Blue Button	Claims/EOB content for Patient Access (ExplanationOfBenefit)

Build status: ORANGE

278 exists but only ~one-third electronic; portal + fax dominate today. FHIR (Da Vinci PAS/CRD/DTR) mandated for CMS-regulated payers Jan 1, 2027. Portal/voice fallback now.

10. Denials

Denials are fully electronic. The 835 ERA (~97% adoption) carries payment and adjustment detail with CARC/RARC codes at both claim and service-line level; the 277CA reports pre-adjudication rejections. From this machine-readable stream, AI can derive denial root cause, recovery probability, correction recommendation and appeal opportunity. Note: 835/ERA requires per-payer enrollment, and denial *data* is electronic even though the resulting *appeal* often is not.

Build status: GREEN

835 is 97% electronic and carries CARC/RARC + 277CA. AI can derive root cause and recovery path. Enrollment required per payer for 835.

11. Appeals

Appeals have **no universal standard or API**. Availability is payer-specific: some payers accept electronic or portal appeals with document upload; many still require fax or mail; and clinical appeals require human authorship. Appeals must therefore be classified per payer (API / Portal / EDI / electronic-non-API / fax / mail / human / unknown) and orchestrated with human review — the platform should never auto-submit an appeal.

Build status: RED

No universal appeal standard or API. Payer-specific: portal, document upload, fax, mail. Human + portal/voice orchestration required.

12. Payer Intelligence

Payer intelligence is the proprietary layer: for each payer, structured operational knowledge of identity and IDs, supported transactions, API/FHIR availability, portal and enrollment requirements, clearinghouse routing, prior-auth and appeal workflows, documentation needs, and — learned over time from outcomes — denial patterns, resolution timing and preferred channel. It is built from permitted, tenant-isolated customer data; PHI from one customer is never shared with another. The per-payer capability matrix (Appendix A) is the seed of this layer.

Figure 4. Channel maturity today by workflow. EDI/clearinghouse is mature for eligibility, benefits, status and denials; direct payer FHIR is emerging (and mandated for PA/provider-access in 2027).

13. API Availability

Across the eight workflows, API availability is high where a HIPAA transaction exists and is intermediated by a clearinghouse, and low where no standard exists (appeals) or adoption is immature (prior auth). The critical nuance for investors: 'API available' almost always means 'available via clearinghouse', not 'the payer publishes a free public API'.

Figure 5. Portal and voice/human dependency by workflow — concentrated in prior authorization, appeals and patient responsibility.

14. Free vs. Paid API Analysis

The investor question — 'are these APIs free?' — must be answered precisely. Free **sandboxes** are common; a free **production** tier is rare. Stedi is the notable self-serve, pay-as-you-go option with a genuine free tier (100 transactions/month) and no monthly minimum; Optum and Availity offer free sandboxes but gate production behind contracts; Waystar is enterprise. Exact per-transaction pricing behind vendor calculators is UNKNOWN and must be captured directly (see Research Gaps).

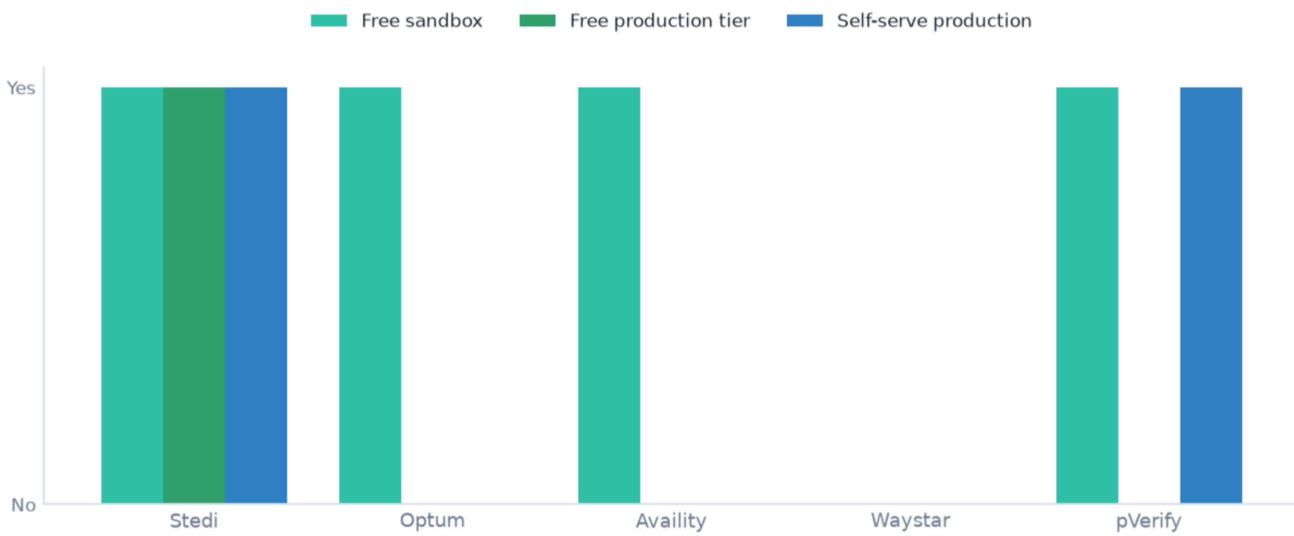

Figure 6. Free sandbox vs. free production vs. self-serve production across the leading vendors.

Do not conflate

A 'free sandbox account' is not a 'free production API'. Record Sandbox = YES, Production = YES, Production Free = NO, Production Pricing = usage-based — as separate fields. Likewise, 'clearinghouse supports payer X' is not 'payer X provides an API'.

15. Clearinghouse / API Infrastructure

The practical path to payer connectivity is the clearinghouse/API layer. The table below summarizes the leading vendors across the transactions an RCM platform needs. Stedi is the strongest fit for an API-first MVP; Optum offers the broadest reach and the rare direct provider-facing RCM API; Availity is strong for Blues and FHIR prior auth; Waystar is enterprise; Zelis is payments/ERA only.

Vendor	Elig	Status	Claims	835	Prior Auth	FHIR	Free Sandbox	Free Prod Tier	Payers
Stedi	YES	YES	YES (P/I/D)	YES	NO / UNKNOWN	NO	YES (free)	YES — 100 free txns/mo (Basic)	3,500+
Availity	YES	YES (276)	YES (I/P/D)	YES	YES (278 + FHIR PA)	YES	YES (Demo — mock only)	Essentials portal free for sponsored payers (not the API)	Blues-heavy ; founded by/for multiple BCBS plans
Optum (Change Healthcare)	YES	YES	YES	YES (+275 a ttachm ents)	UNKNOWN	PARTIA L (Claim FHIR API)	YES (free, no obligation)	NO	"Most U.S. payers" (exact count not public)
Waystar	YES	YES	YES	YES	YES (PA)	UNKNO WN	UNKNOWN / likely none	NO	5,000+ payer connections
Zelis	NO	NO	NO	YES	NO	NO	UNKNOWN	NO	330+ via single ERA/EFT enrollment; 550+ insurers
pVerify	YES	YES	PARTIAL	UNKNO WN	YES	YES	Free trial	NO	Many
Claim.MD	YES (400+ payers)	YES	YES	YES	UNKNOWN	NO	UNKNOWN	UNKNOWN	400+ eligibility
Office Ally	YES	YES	YES	YES	NO	NO	UNKNOWN	Portal free (fees may apply)	Large
Inovalon (ABILITY)	YES (2,300+ payers)	YES	YES	YES	UNKNOWN	PARTIA L (patient access)	YES	NO	2,300+
Edifecs	N/A	N/A	N/A	N/A	N/A	X12/FHI R engine	NO	NO	N/A

Stedi detail: JSON + raw X12 on every transaction; real-time and batch (up to 10,000 eligibility checks/request); 3,500+ payers; free sandbox and a free Basic tier of 100 transactions/month; pay-as-you-go from a $100 balance with no monthly minimum, no per-provider or per-payer fee. It does not currently document 278 or FHIR. Exact per-transaction pricing sits behind an interactive calculator and should be captured before financial modeling.

16. CMS 2027 Interoperability Requirements

Two CMS rules shape the forward trajectory. **CMS-9115-F** (2020) required CMS-regulated payers to build a patient-authorized **Patient Access API** (enforced July 2021) and a public **Provider Directory API** (2021); its payer-to-payer provision was never enforced. **CMS-0057-F** (2024) adds three provider/payer-facing FHIR APIs and enhances Patient Access.

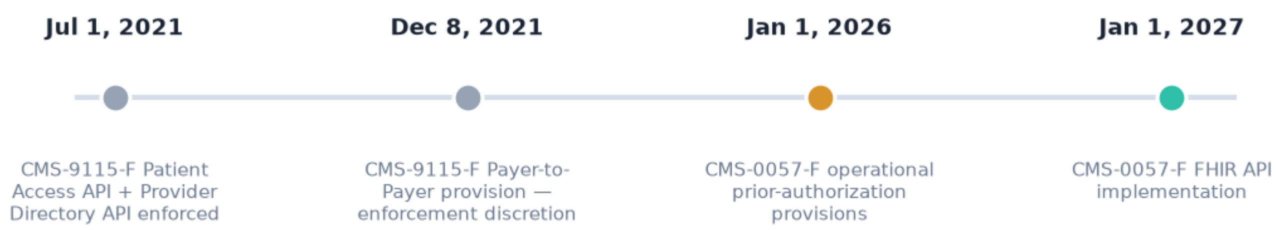

Figure 7. CMS interoperability timeline. Operational prior-auth rules begin Jan 1, 2026; the FHIR API build-out is due Jan 1, 2027.

API (CMS-0057-F)	Data / scope	Notes
Patient Access API	**Enhanced to include prior-authorization information (excl. drugs)**	**Patient-authorized (SMART/OAuth)**
Provider Access API	**Claims/encounter (excl. remittances & cost-sharing), USCDI, prior-auth info**	**In-network providers; attribution/bulk; patient opt-out — the RCM-relevant new API**
Payer-to-Payer API	**Claims/encounter, USCDI, prior-auth info at enrollee direction**	**Replaces the unenforced CMS-9115-F provision**
Prior Authorization API	**End-to-end electronic PA (FHIR; Da Vinci CRD/DTR/PAS recommended)**	**HIPAA enforcement discretion for all-FHIR implementers; X12 278 not banned**

Impacted payers: Medicare Advantage organizations; state Medicaid & CHIP fee-for-service; Medicaid & CHIP managed care; QHP issuers on the Federally-Facilitated Exchanges. The rule does **not** apply to commercial or employer (ERISA) group health plans, which may adopt voluntarily. It mandates FHIR APIs for defined use cases; it does **not** make eligibility API-only and does **not** eliminate X12 (270/271, 278).

> **The RCM-relevant new API**
>
> The **Provider Access API** is the one to track: it shares attributed patients' claims/encounter data (excluding remittances and cost-sharing), USCDI clinical data, and prior-auth information with in-network providers, via bulk/attribution with patient opt-out. For CMS-regulated payers, this is the first broadly-mandated provider-facing payer FHIR feed — live Jan 1, 2027.

17. Payer-by-Payer Capability Matrix

The full 113-field record for all 43 payer organizations is delivered in the accompanying Excel workbook (PAYER MASTER sheet) and CSV. A condensed view is provided in Appendix A. The matrix confirms the pattern: eligibility, claim status and denials are electronically reachable for essentially every payer via clearinghouse; direct provider FHIR is verified only where a developer portal documents it (and is patient-authorized); provider-facing payer FHIR is a 2027 event for impacted payers.

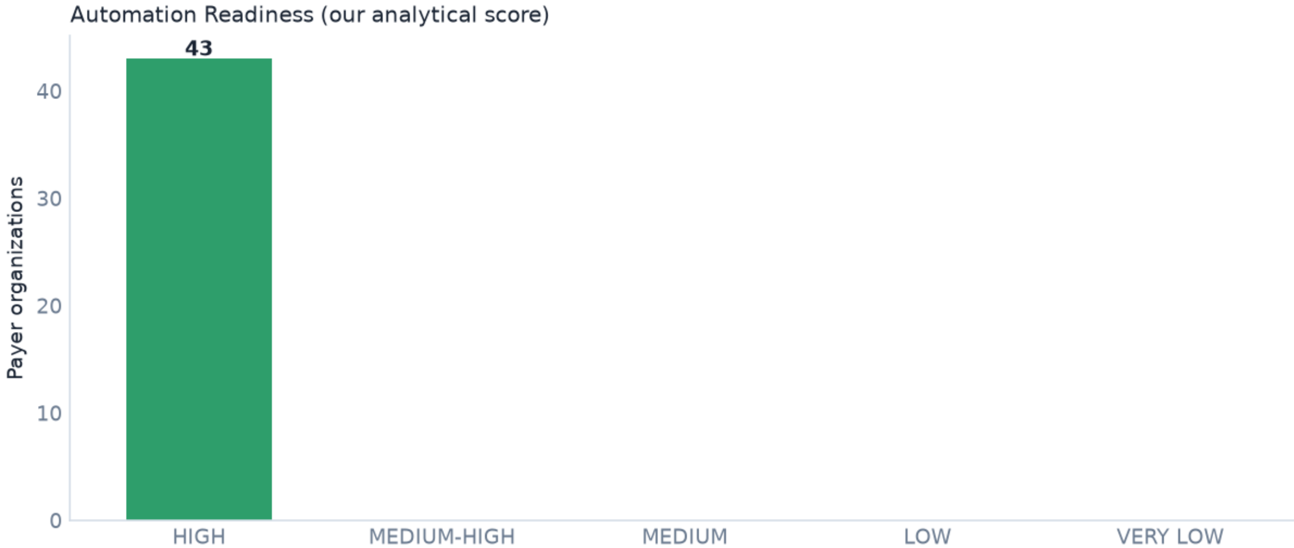

Figure 8. Distribution of payer organizations by Automation Readiness Score band (our analytical score). Most payers are highly automatable for core transactions; prior auth and appeals depress no payer below MEDIUM-HIGH because clearinghouse reach is broad.

18. Recommended MVP Integration Strategy

The hypothesis — use clearinghouse/API infrastructure for V1 and add payer-specific/FHIR selectively, with portal/voice fallback and a long-term orchestration moat — is **validated** by the research. Three options were assessed:

Option	Description	Assessment
1 — Direct payer integrations	Integrate each payer's own APIs	Rejected for V1: direct provider RCM APIs barely exist; enormous per-payer effort for patient-authorized-only FHIR.
2 — Clearinghouse / API infrastructure	Build on Stedi/Optum/Availity	Recommended for V1: one integration reaches thousands of payers for eligibility, benefits, status, 835.
3 — Hybrid	Clearinghouse core + selective payer FHIR + portal/voice	Recommended overall: clearinghouse now, add CMS-0057-F Provider Access/PA FHIR (2027), portal/voice fallback, orchestration layer as the moat.

What we can build today

GREEN — API/EDI feasible now

Eligibility, Benefit verification, Claim status, Denials/835, Claim submission

YELLOW — available, payer-specific / enrollment

Patient responsibility estimation (calc engine), 835 enrollment, some prior-auth requirements (CRD)

ORANGE — portal/manual still required

Prior authorization submission (pre-2027), document upload, payer-specific detail

RED — no reliable standardized API

Appeals (no standard), guaranteed patient liability

Phased recommendation

- **MVP (2–3 capabilities):** Eligibility + Benefit verification + Claim status — via clearinghouse API (Stedi primary), free sandbox → free/low-cost production.
- **V1 (5–6 capabilities):** add Denial intelligence (835/CARC/RARC), Patient-responsibility estimation, and Claim submission/correction (837).
- **Production:** add Prior authorization (portal/voice now → FHIR PAS 2027), Appeals orchestration (portal/voice/human), payer FHIR Provider Access (2027), and the proprietary Payer Intelligence + Orchestration layer.

19. Recommended Production Architecture

The platform is an orchestration layer above existing connectivity. A Payer Intelligence layer selects the cheapest sufficient channel per payer and workflow — EDI → API/FHIR → Portal → Voice → Human — and routes uncertain, high-value or compliance-sensitive cases to human review with a full audit trail.

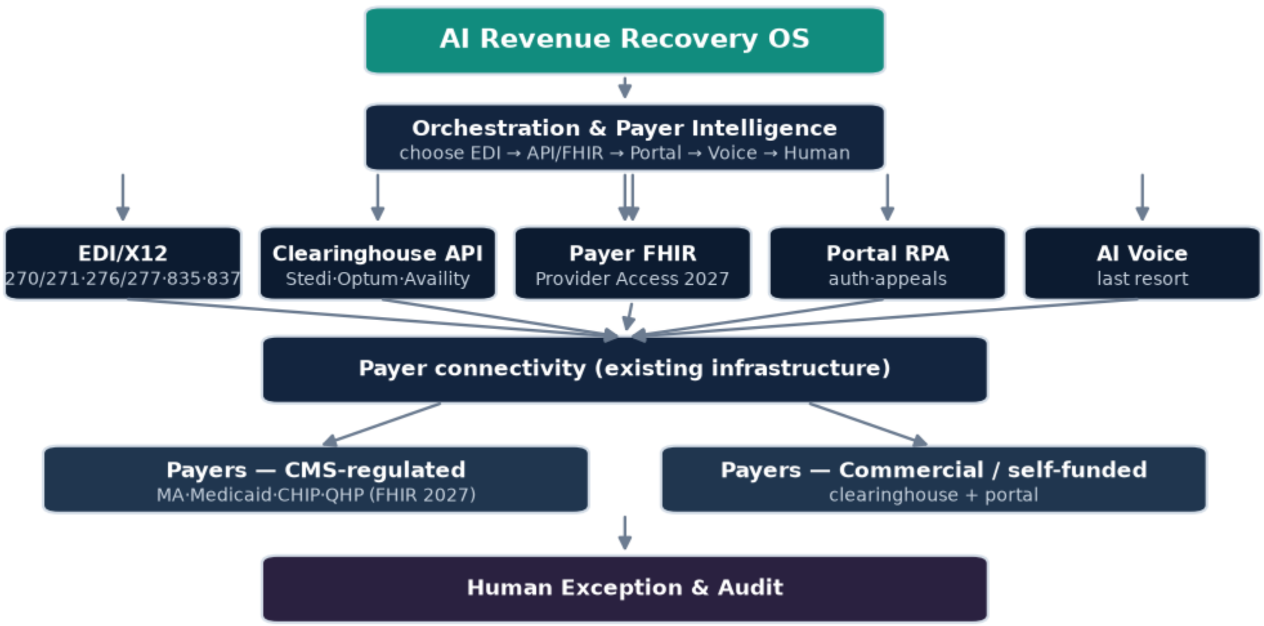

Figure 9. Recommended production architecture — orchestration and payer intelligence over clearinghouse EDI/API, emerging payer FHIR, portal automation and AI voice, with human exception handling.

20. Risks & Limitations

- **Pricing opacity.** Exact per-transaction clearinghouse pricing is behind calculators/contracts (Stedi calculator; Optum/Availity quotes) — model economics only after capturing real numbers.
- **Enrollment friction.** 835/ERA (and some claims/PA) require per-payer, per-transaction enrollment; onboarding time is a real operational cost.
- **Benefit-field variability.** 271 completeness varies by payer; patient-responsibility estimates carry inherent uncertainty and must be labeled estimates.
- **Prior-auth timing.** FHIR PA is a 2027 mandate for CMS-regulated payers only; commercial PA remains portal/fax-heavy for the foreseeable future.
- **Appeals fragmentation.** No standard; per-payer channel discovery and human authorship required.
- **Coverage limitation.** 43 organizations are representative, not the full universe of thousands of payer IDs; several data points remain UNKNOWN pending per-payer verification (Appendix C).
- **Regulatory reliance.** CMS dates and scope are accurate as of research, but implementation maturity at each payer must be verified before depending on a 2027 API.

21. Conclusions

Electronic access to the core revenue-cycle transactions is a solved problem at the standards and infrastructure level, and reachable today through clearinghouse APIs with free sandboxes. The scarce, defensible asset is not connectivity but **intelligence and orchestration**: knowing, per payer and per claim, which channel will resolve the issue at lowest cost, executing across all of them, and learning from every outcome. Prior authorization and appeals remain the hard, human-heavy frontier — and precisely where the 2026–2027 CMS mandates and an AI orchestration layer create the opening.

Bottom line

Do not build a payer network from scratch. Build on existing EDI/clearinghouse/FHIR/portal/voice rails, start with eligibility, benefits and claim status, add 835 denial intelligence and patient-responsibility estimation, and invest the proprietary effort in the Payer Intelligence + Orchestration layer that decides and executes the next best action.

Appendix A. Payer Database (condensed)

Condensed capability view for all 43 organizations. Full 113-field records are in the Excel workbook and CSV.

Payer Organization	Type	Elig	Status	835	Prior Auth	FHIR	Readiness
UnitedHealthcare	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Optum (UHG)	NAT	YES	YES	YES	PARTIAL	UNKNOWN	91 (HIGH)
Aetna	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Cigna Healthcare	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Evernorth	NAT	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
Humana	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Elevance Health (Anthem)	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Centene	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Molina Healthcare	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Kaiser Permanente	NAT	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Health Net	NAT	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
WellCare	MA	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Ambetter	QHP	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Health Care Service Corp (HCSC)	BCBS	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Independence Blue Cross	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Highmark	BCBS	YES	YES	YES	PARTIAL	YES	90 (HIGH)
CareFirst BCBS	BCBS	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Florida Blue	BCBS	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Blue Shield of California	BCBS	YES	YES	YES	PARTIAL	YES	90 (HIGH)
Anthem Blue Cross (CA)	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Premera Blue Cross	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Regence (Cambia)	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
BCBS of Michigan	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Blue Cross NC	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Horizon BCBS NJ	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
BCBS of Tennessee	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Excellus BCBS	BCBS	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
UnitedHealthcare Community Plan	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Aetna Better Health	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Amerigroup / Wellpoint	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)

Payer Organization	Type	Elig	Status	835	Prior Auth	FHIR	Readiness
AmeriHealth Caritas	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
CareSource	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Superior HealthPlan	MCO	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	85 (HIGH)
Oscar Health	QHP	YES	YES	YES	PARTIAL	YES	90 (HIGH)
UMR	TPA	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
Meritain Health	TPA	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
GEHA	TPA	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
Allied Benefit Systems	TPA	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
WebTPA	TPA	YES	YES	YES	PARTIAL	UNKNOWN	83 (HIGH)
Medicare FFS (CMS / MACs)	GOV	YES	YES	YES	PARTIAL	UNKNOWN	82 (HIGH)
Medicaid FFS (State agencies)	GOV	YES	YES	YES	PARTIAL	PARTIAL (CMS-9115)	84 (HIGH)
TRICARE	GOV	YES	YES	YES	PARTIAL	UNKNOWN	82 (HIGH)
Railroad Medicare	GOV	YES	YES	YES	PARTIAL	UNKNOWN	82 (HIGH)

Appendix B. API Sources

Source	Type	Confidence
CMS-0057-F Interoperability & Prior Authorization Final Rule — Fact Sheet	OFFICIAL	High
CMS-0057-F Federal Register (89 FR 8758, doc 2024-00895)	OFFICIAL	High
CMS Prior Authorization API FAQ	OFFICIAL	High
CMS Provider Access API FAQ	OFFICIAL	High
CMS-9115-F Interoperability and Patient Access — Fact Sheet	OFFICIAL	High
CMS-9115-F Payer-to-Payer enforcement discretion FAQ	OFFICIAL	High
CMS Administrative Simplification — Adopted Standards & Operating Rules	OFFICIAL	High
CMS Health Care Transactions Basics	OFFICIAL	High
CMS Operating Rules for Eligibility and Claims Status	OFFICIAL	High
CMS APIs & Implementation Guides (Standards/IGs)	OFFICIAL	High
X12 Claim Adjustment Reason Codes (CARC)	STANDARD	High
X12 Remittance Advice Remark Codes (RARC)	STANDARD	High
CAQH Index Report 2024 — From Transactions to Trust	INDUSTRY	High
CAQH 2024 Index Report — Key Takeaways	INDUSTRY	High
CAQH CORE Eligibility & Benefits (270/271) Data Content Rule	STANDARD	High
CAQH CORE Payment & Remittance (835/EFT) Reassociation Rule	STANDARD	High
CAQH CORE Prior Authorization White Paper (2024)	INDUSTRY	High
HL7 Da Vinci Prior Authorization Support (PAS) IG	STANDARD	High
HL7 Da Vinci Coverage Requirements Discovery (CRD) IG	STANDARD	High
HL7 Da Vinci Documentation Templates and Rules (DTR) IG	STANDARD	High
HL7 Da Vinci Clinical Data Exchange (CDex) IG	STANDARD	High
HL7 Da Vinci Payer Data Exchange (PDex) IG	STANDARD	High
CARIN Alliance IG for Blue Button (CARIN-BB)	STANDARD	High
Stedi Healthcare docs	VENDOR	High
Stedi pricing (pay-as-you-go)	VENDOR	High
Stedi Basic (free) plan — 100 free transactions/mo	VENDOR	High
Stedi pay-as-you-go pricing announcement	VENDOR	High
Stedi payer network (3,500+ payers)	VENDOR	High
Stedi transaction enrollment	VENDOR	High
Availity API Marketplace	VENDOR	High
Availity HIPAA Transactions (developer blog)	VENDOR	High
Availity developer getting started (Demo vs Standard plan)	VENDOR	High
Optum (Change Healthcare) developer portal	VENDOR	High
Optum Eligibility & Claims API — get started	VENDOR	High
Waystar platform — claim management	VENDOR	Medium
Zelis provider payments / ERA	VENDOR	High
pVerify eligibility/claim-status/PA API (FHIR)	VENDOR	Medium
Claim.MD clearinghouse API	VENDOR	Medium
Office Ally EDI clearinghouse	VENDOR	Medium
Inovalon (ABILITY) provider cloud / ONE developer portal	VENDOR	Medium

Source	Type	Confidence
ONC Lantern FHIR endpoint monitoring	OFFICIAL	High
CARIN Alliance FHIR Directory Framework	INDUSTRY	Medium

Full URLs are hyperlinked in the Excel SOURCES sheet. Primary sources: CMS (cms.gov, Federal Register 2024-00895 / 2020-05050), ASC X12 (x12.org), HL7 (hl7.org Da Vinci & CARIN IGs), CAQH (CORE rules, CAQH Index 2024), and vendor developer portals (Stedi, Optum/Change, Availity, Waystar, Zelis).

Appendix C. Glossary

Term	Definition
270/271	Eligibility/benefit inquiry and response (X12).
276/277	Claim status request and response (X12).
277CA	Claim acknowledgment — pre-adjudication accept/reject of an 837.
278	Health care services review — referral/prior-authorization (X12).
835	Electronic remittance advice (ERA) — payment/adjustment detail.
837	Electronic health care claim (P=professional, I=institutional, D=dental).
CARC	Claim Adjustment Reason Code — why a claim/line was adjusted (X12-maintained).
RARC	Remittance Advice Remark Code — supplemental remittance detail (CMS-maintained).
CAQH CORE	Operating rules for eligibility/claim-status content, infrastructure & connectivity.
FHIR	HL7 Fast Healthcare Interoperability Resources (R4 = 4.0.1).
SMART on FHIR	OAuth2/OpenID authorization framework for FHIR apps.
Patient Access API	CMS-9115-F FHIR API returning a member's own data with the member's authorization.
Provider Access API	CMS-0057-F FHIR API sharing attributed-patient data with in-network providers (2027).
Da Vinci	HL7 accelerator publishing payer/provider FHIR IGs (CRD, DTR, PAS, CDex, PDex).
Clearinghouse	Intermediary that translates and routes X12 transactions between providers and payers.
ERA/EFT reassociation	Matching an 835 to its EFT deposit via the TRN trace number (Nacha CCD+).
CMS-0057-F	2024 CMS Interoperability & Prior Authorization Final Rule.
CMS-9115-F	2020 CMS Interoperability & Patient Access Final Rule.

Research coverage limitation. This report is a feasibility analysis compiled from public documentation. Where complete payer-level verification was not possible, values are marked UNKNOWN and enumerated in the Research Gaps appendix of the Excel workbook. Nothing herein should be treated as a production commitment without per-payer verification and confirmation of current clearinghouse pricing.